

A Proof of the Power Rule using Pascal's Triangle

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1 Claim

$$\frac{d}{dx}x^n = nx^{n-1} \quad x \in \mathbb{R}, n \neq 0$$

2 Proof

Begin with the limit definition of the derivative for x^n , which is as follows:

$$\frac{d}{dx}x^n = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

Notice that the coefficient of the k th term in the expansion of $(x+h)^n$ will be the k th number in row n of Pascal's triangle, denoted $P_{n,k}$. For example, the 3rd coefficient of $(x+h)^3$ will be $P_{3,3} = 3$. Performing the expansion, we obtain

$$(x+h)^3 = x^3 + 3x^2h + 3h^2x + h^3$$

An by inspection we can see that the third coefficient is 3, as we expect. Using the general formula for Pascal's triangle, we have

$$(x+h)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} h^k$$

Substituting this into the difference quotient, we have

$$\lim_{h \rightarrow 0} \frac{x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots h^n - x^n}{h}$$

Which we can simplify to

$$\lim_{h \rightarrow 0} \left(nx^{n-1} + \binom{n}{2}x^{n-2}h + \dots + h^{n-1} \right)$$

By direct substitution with $h = 0$, all the terms except the first cancel, and we are left with

$$\frac{d}{dx}x^n = nx^{n-1}$$

QED